

Hexadezimal system

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$$1) \underbrace{0110}_6 \underbrace{1110}_E_2 = 6E_{16} \quad 2) \underbrace{1110}_E \underbrace{1111}_F_2 = EF_{16}$$

$$3) \underbrace{0100}_4 \underbrace{1010}_A \underbrace{1001}_9 \underbrace{1011}_B_2 = 4A9B_{16}$$

$$4) \underbrace{1101}_D, \underbrace{0101}_5 \underbrace{1100}_C_2 = D,5C_{16}$$

$$5) \underbrace{1101}_D, \underbrace{1111}_F \underbrace{1011}_B_2 = D,FB_{16}$$

$$6) \underbrace{1011}_B, \underbrace{0001}_1 \underbrace{1010}_A \underbrace{0001}_1_2 = B,1A1_{16}$$

$$7) \begin{array}{ccc} B & A & D_{16} \\ 1011 & 1010 & 1101_2 \end{array}$$

$$8) \begin{array}{cccc} B & A & 5 & 5_{16} \\ 1011 & 1010 & 0101 & 0101_2 \end{array}$$

$$9) \begin{array}{cccc} A & F & F & E_{16} \\ 1010 & 1111 & 1111 & 1110_2 \end{array}$$

$$10) \begin{array}{ccc} C & 0, & 01_{16} \\ 1100 & 0000, & 0000 \ 0001_2 \end{array}$$

$$11) \begin{array}{cccccc} B & A & D, C & A & F & E_{16} \\ 1011 & 1010 & 1101, 1100 & 1100 & 1111 & 1110_2 \end{array}$$

$$12) \begin{array}{cccccc} B & A & D, A & 5 & 5_{16} \\ 1011 & 1010 & 1101, 1010 & 0101 & 0101_2 \end{array}$$

$$13) \begin{array}{l} 42_{10} : 16 = 2 \quad 10R \rightarrow A \\ 2 : 16 = 0 \quad 2R \rightarrow 2 \end{array}$$

$$42_{10} = 2A_{16}$$

$$14) \begin{array}{l} 4711_{10} : 16 = 294 \quad 7R \\ 294 : 16 = 18 \quad 6R \\ 18 : 16 = 1 \quad 2R \\ 1 : 16 = 0 \quad 1R \end{array}$$

$$4711_{10} = 1267_{16}$$

15)

$$\begin{aligned}
 12345678 : 16 &= 771604 \\
 771604 : 16 &= 48225 \\
 48225 : 16 &= 3014 \\
 3014 : 16 &= 188 \\
 188 : 16 &= 11 \\
 11 : 16 &= 0
 \end{aligned}$$

$$\begin{aligned}
 14R &\rightarrow E \\
 4R & \\
 7R & \\
 6R & \\
 12R &\rightarrow C \\
 11R &\rightarrow B
 \end{aligned}$$

$$\Rightarrow 12345678_{10} = BC614E_{16}$$

16)

$$2_{10} : 16 = 0 \quad 2R$$

$$\Rightarrow 2,66796875_{10} = 2,AD_{16}$$

$$\begin{aligned}
 0,66796875 \cdot 16 &= 10,6875 \\
 0,6875 \cdot 16 &= 11
 \end{aligned}$$

$$\begin{aligned}
 &\rightarrow 10 \rightarrow A \\
 &\rightarrow 11 \rightarrow B
 \end{aligned}$$

17)

$$\begin{aligned}
 192 : 16 &= 12 \quad OR \\
 12 : 16 &= 0 \quad 12R \rightarrow C
 \end{aligned}$$

$$\Rightarrow 192,125_{10} = C0,2_{16}$$

$$0,125 \cdot 16 = 2 \rightarrow 2$$

18)

$$11 : 16 = 0 \quad 11R \rightarrow B$$

$$11,22_{10} = \overline{B,3857ED8}$$

$$0,22 \cdot 16 = 3,52 \rightarrow 3$$

$$0,52 \cdot 16 = 8,32 \rightarrow 8$$

$$0,32 \cdot 16 = 5,12 \rightarrow 5$$

$$0,12 \cdot 16 = 1,92 \rightarrow 1$$

$$0,92 \cdot 16 = 14,72 \rightarrow 14 \rightarrow E$$

$$0,72 \cdot 16 = 11,52 \rightarrow 11 \rightarrow B$$

$$0,52 \cdot 16 = 8,32 \rightarrow 8$$

:

19)

$$E : 14 \cdot 16^0 = 14$$

$$D : 13 \cdot 16^1 = 208$$

$$0 : 0 \cdot 16^2 = 0$$

$$C : 12 \cdot 16^3 = \underline{49152}$$

$$49374$$

$$CODE_{16} = 49374_{10}$$

20)

$$E : 14 \cdot 16^0 = 14$$

$$5 : 5 \cdot 16^1 = 80$$

$$A : 10 \cdot 16^2 = 2560$$

$$B : 11 \cdot 16^3 = \underline{45056}$$

$$\underline{47710}$$

$$BASE_{16} = 47710_{10}$$

21)

$$\begin{aligned}
 D &: 13 \cdot 16^0 = 13 \\
 0 &: 0 \cdot 16^1 = 0 \\
 0 &: 0 \cdot 16^2 = 0 \\
 F &: 15 \cdot 16^3 = 61440 \\
 D &: 13 \cdot 16^4 = 851968 \\
 1 &: 1 \cdot 16^5 = 1048576 \\
 0 &: 0 \cdot 16^6 = 0 \\
 C &: 12 \cdot 16^7 = 3221225472 \\
 &\quad \underline{3221225472}
 \end{aligned}$$

$$C01DFOOD_{16} = 3223187469_{10}$$

22)

$$\begin{aligned}
 D &: 13 \cdot 16^0 = 13 \\
 1 &: 1 \cdot 16^1 = 16 \\
 &\quad \underline{29}
 \end{aligned}$$

$$1: 1 \cdot 16^0 = 1$$

$$A: 1 \cdot 16^1 = 16$$

$$E: 1 \cdot 16^2 = 3584$$

$$\Rightarrow \frac{3745}{16^3} = 0,914306640625$$

$$1D, EA_{16} = 29,914306640625_{10}$$

23)

$$\begin{aligned}
 A &: 10 \cdot 16^0 = 10 \\
 C &: 12 \cdot 16^1 = 192 \\
 0 &: 0 \cdot 16^2 = 0 \\
 C &: 12 \cdot 16^3 = 49152 \\
 &\quad \underline{49354}
 \end{aligned}$$

$$A: 10 \cdot 16^0 = 10$$

$$1: 1 \cdot 16^1 = 16$$

$$0: 0 \cdot 16^2 = 0$$

$$C: 12 \cdot 16^3 = 49152$$

$$\Rightarrow \frac{49178}{16^4} =$$

$$= 0,750396728515625$$

$$C0CA, CD1A_{16} = 49354,750396728515625_{10}$$

24)

$$\begin{aligned}
 D &: 13 \cdot 16^0 = 13 \\
 A &: 10 \cdot 16^1 = 160 \\
 E &: 14 \cdot 16^2 = 3584 \\
 D &: 13 \cdot 16^3 = 52248 \\
 &\quad \underline{57005}
 \end{aligned}$$

$$F: 15 \cdot 16^0 = 15$$

$$E: 14 \cdot 16^1 = 224$$

$$E: 14 \cdot 16^2 = 3584$$

$$D: 13 \cdot 16^3 = 45056$$

$$\Rightarrow \frac{48829}{16^4} =$$

$$= 0,7458343505859375$$

$$DEAD, BEEF_{16} = 57005,7458343505859375_{10}$$